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No. | <p>Note: Start answer of a fresh question from fresh page only. Direct answer to a question will not be entertained.</p> | course
outcome
(CO)
CO301.1 |
| 1. | A hash function defined $ash(key) = key \bmod 5$, with linear probing, is used to insert the keys 134, 159, 125, 18, 143 one by one into an initially empty table indexed from 0 to 5. What will be the location of key 143? | CO301.1 |
| 2. | Consider a double hashing using the two hash functions: $hash(key) = key \bmod m$ and $hash2(key) = key \bmod s$, where $hash2(key)$ is used only when there is a collision. For collision avoidance, the combined double-hash function $(hash(key) + i * hash2(key)) \% m$, where i is initialized to 1 and increases by 1 whenever $hash2(key)$ is used. Assuming an initially empty hash table, with size $m = 7$, $s = 3$ and index starting from 0, which of the following is the contents of the table after the sequence of numbers 64, 12, 13, 9, and 50 are inserted one by one into the table? (Note: '_' denotes an empty location in the table). | CO301.1 |
| 3. | Given the following input: 55, 775, 218, 168, 45 and the hash function $h = key \bmod 5$. If chaining is used for collision resolution, and the above elements are inserted one by one into an initially empty table. Find the index containing maximum elements chained. | CO301.1 |
| 4. | Consider a hash table that uses the linear probing technique with the following hash function $f(x) = (5x+4)\%11$. (The hash table is of size 11.) If we insert the values 3, 9, 2, 1, 14, 6 and 25 into the table, in that order, show where these values would end up in the table? | CO301.1 |